Supporting information for *The Selling closure: a complete operator set for reindexing, and the price of pseudo-symmetry*

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Table, section and equation numbers of the main text are cited as there. Conorm vectors are written in the slot order da, db, dc, ab, ac, bc of the main text, and strains in per cent. Section S6 lists, for every section, table and figure, the check that reproduces its numbers.

S1. Catalogues of shapes and specialisations

Cell shapes. Table S1 lists the shapes of the ten sorts with more than one, each as the cell with the most right angles and no acute angle. The three shapes of a primitive monoclinic lattice are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16); the extra shapes of primitive orthorhombic, tetragonal and cubic lattices are monoclinic-looking cells with two right angles; C-centred orthorhombic has two primitive shapes; the boundary sorts of body-centred orthorhombic, C-centred monoclinic and triclinic have two or three cells with a single right angle each, for triclinic the known instability of the reduced cell when an angle is near 90° (Andrews *et al.*, 1980).

Table S1. *The cell shapes of the sorts with more than one, for the cells named; each shape is given as the cell with the most right angles and all angles* $\geq 90°$*, shortest edges first when several qualify and b unique for the monoclinic-looking ones, with the number n of members (up to swaps) that have it. Centred cells are given by their primitive cells; the boundary sorts are specified by their conorms in* $Å^2$*.*

sort	lattice	n	shape
cP	30 Å	4	30, 30, 30; 90, 90, 90
		12	30, 30, 42.4; 90, 135, 90
tP	30, 30, 50 Å	4	30, 30, 50; 90, 90, 90
		4	30, 30, 58.3; 90, 121.0, 90
		8	30, 50, 42.4; 90, 135, 90
oP	30, 40, 55 Å	4	30, 40, 55; 90, 90, 90
		4	30, 40, 62.6; 90, 118.6, 90
		4	30, 55, 50; 90, 126.9, 90
		4	40, 55, 50; 90, 143.1, 90
oS	C 30, 40, 50 Å	2	25, 50, 25; 90, 106.3, 90
		4	25, 50, 30; 90, 126.9, 90
mP	30, 40, 50 Å, 100°	2	30, 40, 50; 90, 100, 90
		2	30, 40, 53.7; 90, 113.4, 90
		2	50, 40, 53.7; 90, 146.6, 90
oI (O3)	900, 1600, 0, 0, 1600, 900	1	50, 50, 50; 111.1, 129.8, 90
		2	42.4, 50, 50; 90, 115.1, 115.1
mC (M2B)	900, 1600, 2500, 0, 1600, 900	1	50, 50, 70.7; 104.7, 116.9, 90
		1	50, 50, 65.6; 105.9, 105.9, 90
mC (M3)	2500, 1600, 0, 0, 1600, 900	1	50, 50, 64.0; 120, 90, 111.1
		2	50, 56.6, 58.3; 90, 108.0, 124.4
aP (A2)	2500, 400, 1200, 0, 1600, 900	1	36.1, 60.8, 64.0; 114.3, 90, 114.2
		1	36.1, 56.6, 64.0; 116.2, 90, 101.3
aP (A3)	2500, 400, 0, 0, 1600, 900	1	36.1, 50, 53.9; 90, 101.9, 119.9
		1	44.7, 50, 53.9; 90, 99.6, 135.7
		1	36.1, 44.7, 58.3; 90, 115.3, 104.4

Specialisations. The conorms of any member, in any labelling, are integer combinations of the reduced cell's conorms, hence linear in the free parameters of the sort (the letters of Table 6 of the main text). Imposing another sort's zeros and equalities on one member is a homogeneous linear system in those parameters, and a solution with non-negative parameters, a non-singular lattice and a resulting lattice of the target sort and no higher is a specialisation. Table S2 lists the conditions of codimension one, on 36 pairs of sorts. Each is a parameter vanishing or two parameters becoming equal. The classical values appear: body-centred tetragonal is face-centred cubic at $c/a = \sqrt{2}$ and body-centred cubic at $c/a = 1$; C-centred orthorhombic is hexagonal at $b = \sqrt{3}\,a$ and tetragonal at $b = a$; rhombohedral is cubic at $\alpha = 60°$, $90°$ and

109.47°. The monoclinic conditions come in threes, one per shape: with unique axis c and plane lattice a, b, the lattice is C-centred orthorhombic when $|a| = |b|$, $|a| = |a+b|$ or $|b| = |a+b|$, and primitive orthorhombic when $a \perp b$, $a \perp (a+b)$ or $b \perp (a+b)$ — the textbook condition $a = c$ read in each of the three cell choices. The table also shows which sorts are one condition from a much higher symmetry: C-centred monoclinic from tetragonal, body-centred tetragonal and rhombohedral; the body-centred orthorhombic boundary sort from face-centred cubic; triclinic on a wall from C-centred monoclinic.

Table S2. *All specialisations of codimension one between the 24 sorts. Conditions are on the letters of the sort's pattern in Table 6 of the main text; "shapes" is the number of distinct cell shapes of the starting sort through which the conditions are reached, and the last column reads the condition in the cell of the conventional description where that is short.*

from	to	conditions	shapes	reading
aP A1	aP A2	any letter $= 0$	1	a right angle appears
aP A2	aP A3	$v = 0$	1	a second right angle
aP A2	mC M2A	$s = t$, $r = u$	1	
aP A2	mP M4	r, s, t or $u = 0$	1	
aP A3	mC M3	any two letters equal	3	one per shape
mP M4	oP O5	$r = 0$, $s = 0$, $t = 0$	3	$a \perp b$, $a \perp (a + b)$, $b \perp (a + b)$
mP M4	oS O4	$s = t$, $r = t$, $r = s$	3	$\lvert a\rvert = \lvert b\rvert$, $\lvert a\rvert = \lvert a+b\rvert$, $\lvert b\rvert = \lvert a+b\rvert$
mC M1A	oF O1A	$r = t$	1	
mC M1A	oI O2	$u = 0$	1	
mC M1A	oS O4	$r = 0$, $t = 0$	1	
mC M1B	mC M2B	$t = 0$, $u = 0$	1	boundary sort
mC M1B	mC M3	$r = 0$, $s = 0$	1	boundary sort
mC M1B	oF O1A	$r = s$	1	
mC M1B	oI O1B	$t = u$	1	
mC M2A	oI O2	$r = t$	1	
mC M2B	oI O3	$t = 0$	1	
mC M2B	tI T2	$r = s$	1	one step to tI
mC M2B	tP T5	$r = 0$, $s = 0$	1	one step to tP
mC M3	oI O3	$r = t$	1	
mC M3	hR R3	$r = s$, $s = t$	1	one step to hR
oP O5	tP T5	$r = s$, $r = t$, $s = t$	1	$a = b$, $a = c$, $b = c$
oS O4	tP T5	$r = 0$	1	$b = a$
oS O4	hP H4	$r = s$	1	$b = \sqrt{3}\, a$
oI O1B	oI O3	r, s or $t = 0$	1	boundary sort
oI O1B	tI T1	$r = s$, $r = t$, $s = t$	1	two edges equal
oI O2	tI T2	$r = s$	1	
oI O3	cF C3	$r = s$	1	one step to cF
oF O1A	tI T1	$s = t$	1	
oF O1A	tI T2	$s = 0$, $t = 0$	1	
tP T5	cP C5	$r = s$	1	$c = a$
tI T1	cI C1	$r = s$	1	$c/a = 1$
tI T1	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from below
tI T2	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from above
hR R1	cP C5	$r = 0$	1	$\alpha = 90^\circ$
hR R1	cI C1	$r = s$	1	$\alpha = 109.47^\circ$
hR R3	cF C3	$r = s$	1	$\alpha = 60^\circ$

S2. The benchmark, the comparisons and the walks

The benchmark. The directory `manuscript/benchmark` of the repository holds 780 measured cells with exact references and true operators, in five families: 75 cells measured in a different shape of the same lattice; 660 lattices placed 0.5 to 8 times the error from a condition of Table S2; 20 cells whose only reduced superbase is a difference-wall

member; 16 needles and plates with angular errors; 9 cells of the boundary sorts. Each case carries its Niggli cell. The errors are isotropic in the metric tensor, of median strain 0.1%. A case counts as found when the true proper operator itself is among the candidates. `evaluate_strain.py` scores the closure and the $\{-1, 0, 1\}$ search at the band and at three times each case's error (`results_strain.txt`); `evaluate.py --niggli` tallies the Niggli cells by family and sort (`results_niggli.txt`). The measured Niggli cell is a different cell from the reference's in 45 cases (O1B 23, O3 12, M1B 6, one each at M2B, M3, R3 and T2; 14 of them a change between type I and type II), and the same cell in another basis in 125 (O1B 25, M1A 15, O1A 15, T1 15, M1B 13, O4 9, O2 7, R1 7, M2A 6, T2 6, O3 5, A2 1, A3 1).

The comparisons. `compare_methods.py` runs the methods of Table 10 of the main text on the benchmark (`results_compare.txt`) and, with `--stress`, on the stress family (`results_stress.txt`): 60 references drawn from the benchmark, each with independent traceless errors on the reference and the new cell, scaled so that the true operator's price is t times the band, $t = 0.5$, 0.9, 1.1 and 1.5, and a random proper reindexing with entries in $\{-2, \ldots, 2\}$. Each method's candidates are made proper, priced by the geodesic distance after the best scale, and kept within the band; at $t = 1.1$ and 1.5 no method found the true operator, as by construction none should. `window_failures.py` builds the window family (`results_window_failures.txt`): 300 lattices of the sorts M4, O4, H4, O5, T5 and C5 with random parameters, each measured twice with an error of 0.1 to 2%, the reference given as its Niggli cell and the new cell in a random proper basis, classified by the Niggli type of the reference and by the kind of face (with an exact reference in the labelling of the sort's pattern the zeros never sit on d and no window fails), and 60 lattices on random two- and three-zero faces whose closure has a member needing an entry of 3, measured in that member. `coset_rank.py` groups the closure's candidates by the exact

symmetries of the underlying lattice (`results_coset.txt`). `niggli_types.py` reduces near-monoclinic cells from their Niggli cells and records where the near-zero conorms land (`results_niggli_types.txt`).

The PDB. `pdb_nonisomorphism.py` measures the price between entries of one crystal form (`results_pdb_nonisomorphism.txt`). `pdb_lepage.py` runs the cctbx lattice-symmetry analysis at 3° on every entry and prices the pseudo-symmetries (`pdb_lepage.jsonl`, `results_pdb_lepage.txt`); `pdb_pricing_summary.py` gives the transitions and the fractions within the non-isomorphism band (`results_pdb_pricing.txt`). Table S8 lists the prices by the Delone sort of the Le Page lattice and Table 11 of the main text by transition from the reported type.

Walks. Table S3 compares four walks on the measured metric, with a band τ on the conorms, and the construction of Section S4 against brute force.

Table S3. *Four walks on the measured metric, and the construction of Section S4, against brute force on 80 cells with zero conorms at awkward positions (orthorhombic, cubic, plate, monoclinic and hexagonal zero patterns), with a band τ on the conorms at* 0.1 *and* 0.5 *of* $\lambda_1/32$.

procedure	cells with a wrong set
flip wherever $\lvert p\rvert \le \tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le \tau$, filter at the end	13
flip wherever $\lvert p\rvert \le 2\tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le 2\tau$, filter at the end	2
idealise at 4τ, enumerate the exact closure, filter	0

S3. The operators alone

The closure can be described without any lattice. An operator $\mathbf{W}$, written as the new basis in the coordinates of the reduced cell, acts on the six conorms by a fixed integer 6×6 matrix $\mathbf{A_W}$, since every conorm of $\mathbf{W}F_0$ is an integer combination of the conorms of F_0. The reduced cells are the points of the orthant $\{p \ge 0\}$ in conorm space. Selling (1874) showed that every lattice has a conorm vector in this orthant, unique up to

the 48 swaps except on the faces (Delaunay, 1933; Conway & Sloane, 1992), and the closure of a lattice is the set of points of its orbit that lie in the orthant: one in the interior, several on the faces. So $\mathbf{W}$ is an operator of the closure of every lattice on the face of Z exactly when $\mathbf{A_W}$ maps the face $\{p \geq 0,\ p_Z = 0\}$ into the orthant, a sign condition on the entries of $\mathbf{A_W}$ and nothing else.

Enumerating integer matrices with entries in $[-2, 2]$ under that condition, over all 38 faces, and closing the result under relabelling at both ends, gives a set $\mathcal{U}$ of $6096 = 48 \times 127$ integer matrices with entries in $\{-3, \ldots, 3\}$; per face there are 48×1, 2, 3, 6 and 16 matrices for no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector and three zeros. Both placements of three zeros, the triangle and the path, were enumerated; they describe the same lattices read from an orthogonal cell or from a cell with a face diagonal, each has 16 members, and no swap carries one placement into the other. $\mathcal{U}$ is closed under inversion, and every element has a labelling whose basis vectors have entries in $[-2, 2]$. That $\mathcal{U}$ contains every operator between two reduced cells of one lattice, whatever the lattice, does not rest on that window: Lemma A1 of the main text confines the members of every closure to superbase coordinates $s + 2k$ with $|k_i| \leq 3$ relative to a reduced cell, and the enumeration of that whole window, over all 38 faces, finds only members that lie in $\mathcal{U}$ in each of their 48 labellings. In the basis of the reduced cell their entries are in $\{-1, 0, 1\}$ when the zero conorms avoid d, and up to 3 otherwise (Table 7 of the main text). Up to relabelling at both ends $\mathcal{U}$ has nine kinds of element, of flip length 0 to 4 (Table S4).

Table S4. *The nine kinds of operator between Selling-reduced cells, up to relabelling at both ends; "matrices" counts the integer matrices of the kind in $\mathcal{U}$, "flips" its length as a flip word, and the last column the zero patterns in whose closures it occurs.*

kind	matrices	flips	occurs with
1	48	0	every pattern (the swaps)
2	288	1	every pattern with a zero
3	1152	2	two zeros sharing a vector; three zeros
4	1152	3	three zeros
5	1152	3	three zeros
6	576	3	two zeros sharing a vector; three zeros
7	576	4	three zeros
8	576	4	three zeros
9	576	4	three zeros

Why four. For a lattice with $a \perp b \perp c$ the sixteen members are four orthogonal superbases, differing by the sign of one axis, and twelve with a face diagonal. A flip from an orthogonal cell always gives a cell with a face diagonal, so changing the sign of one axis takes three flips, out, across and in, and the diagonal cells attached only to the far orthogonal cell take one more: from (a, b, c) the members lie at flip distances $0, 1, 2, 3, 4$ in numbers $1, 3, 6, 3, 3$, and four is the diameter of the largest face.

S4. A band on the conorms

The simplest widening of the closure is an absolute band τ in Å^2 on the conorms,

$$\mathcal{S}_\tau(\mathbf{G}) = \big\{\, F \;:\; \text{all six conorms of } F \text{ under } \mathbf{G} \text{ are } \geq -\tau \,\big\}, \tag{1}$$

Selling reduction with its boundary moved from 0 to $-\tau$. This is an L^∞ band on the conorms, a different norm from the strain of the main text; the two are comparable only for cells whose edges are similar in length. It can be computed by idealising: set every conorm of the reduced superbase F_0 that is at most 4τ to zero, enumerate the exact closure of the idealised lattice Λ^*, and keep the members whose conorms under the measured metric are all at least $-\tau$. The construction is exact when λ_1, the squared length of the shortest lattice vector, exceeds 32τ (Theorem S4.5), a ratio independent of the scale of the cell: for an isotropic cell it allows an angular error

of about 1.8°. It fails for anisotropic cells, where the band must cover the conorms between the long vectors: on the 16 needles and plates of the benchmark it declined 8. On lattices of five zero patterns with zero conorms placed at random it was exact at every band from $\lambda_1/32$ to $\lambda_1/6$ in which the idealised lattice existed; on dense two-zero grids idealising at 2τ was already exact, at τ not. For the worked cell at $\tau = 20\,\text{Å}^2$, within the hypothesis, it returns the six members of the ring. A new cell constructed so that its only reduced superbase is a difference-wall member of an orthorhombic corner was matched against two stored sets of the reference: the complete set offers three candidates and their negatives; a set built by idealising only at the band has lost the third, $\pm\mathbf{I}$, the true operator.

Bands per slot. A band that scales with the vectors of each superbase, $p_{ij}(F) \geq -\sigma|u_i||u_j|$ on every slot of F (all pairwise angles at least $90° - \arcsin\sigma$), is the natural angular form of the L^∞ band, but the same construction applied with each slot of F_0 idealised against its own band is not exact. On the needle $5.58 \times 5.21 \times 131.97\,\text{Å}$, $\gamma = 91.12°$, at $\lambda_1/(\sigma l_a l_c) = 33$ and 40, it returns 6 superbases where brute force finds 12: the conorm ab survives idealisation because its own band is small, while the members built on $a + c$ carry the same dot product on a slot whose band is about 25 times larger. Idealising until the zeroed slots are consistent across the members of the closure repairs this, and on 240 random needles, plates, isotropic and monoclinic cells it is never wrong (exact in 124, refused in 116), while the single-pass construction is wrong in 9 (z57).

For a superbase F and two metrics, $\kappa_F(\mathbf{G}, \mathbf{G}')$ is the largest change among F's six conorms.

Proposition S4.1 (characterisation superbase by superbase). $\mathcal{S}_\tau(\mathbf{G}) = \{F : F \in \mathcal{C}(\mathbf{G}')$ for some $\mathbf{G}'$ with $\kappa_F(\mathbf{G}, \mathbf{G}') \leq \tau\}$. *Proof.* ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\kappa_F \leq \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) The metric whose conorms in the

frame of F are $\max(p_{ij},0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite and F is obtuse for it. □

Lemma S4.2 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\le \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \ge 4m^*$ in some class. *Proof.* In the frame of F_0, Lemma A2 of the main text gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, F is obtuse for Λ^*; otherwise some $\gamma_{ij} \ge 1$ multiplies a $p^0_{ij} \ge m^*$. □

Lemma S4.3 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \le 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \le 8\tau$ has a partial sum of squared length at most $32\tau < \lambda_1$. Lemma A2 in the frame of F gives $0 \le e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \le \tau\sum_N \gamma$. If a large γ were positive, Lemma S4.4(ii) would give $8\tau\sum_{\text{large}}\gamma < \tau\sum_N\gamma \le 8\tau\sum_{\text{large}}\gamma$; so the large γ vanish, and Lemma S4.4(i) gives $e_c \le 4\tau\cdot 4$. □

Lemma S4.4 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \le 4$. (ii) If $\sum_L \gamma_{ij} > 0$, then $\sum_N \gamma_{ij} \le 8\sum_L \gamma_{ij}$. Both bounds are attained. *Proof.* In (i) $|\Delta k| \le 1$ on L, so all $|k_i| \le 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \le 15$; beyond, some $|k_i| \ge 16$, and since L connects the four vertices with $k_0 = 0$, $Q = \sum_L \Delta k^2 \ge 16^2/3 > 75$. For each pattern (L,N) let $\rho_{L,N}$ be the largest generalised eigenvalue of the quadratic form $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$; the largest over all pairs (L,N) is $3+2\sqrt{2}$. Then $\sum_N\gamma \le \rho_{L,N}Q + \sqrt{|N|\rho_{L,N}Q}$ and $\sum_L\gamma \ge Q - \sqrt{|L|Q}$, and the ratio of these bounds, $(\rho_{L,N}\sqrt{Q} + \sqrt{|N|\rho_{L,N}})/(\sqrt{Q} - \sqrt{|L|})$, decreases in Q,

so it suffices to check it at $Q = 75$ for each pattern with its own $\rho_{L,N}$, where it is below 8 in every case; with the uniform value $3 + 2\sqrt{2}$ it would not be. The lemma is computer-assisted. Equality in (ii): $L = \{01, 02, 13\}$, $N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$. □

Theorem S4.5. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) = \{F \in \mathcal{C}(\Lambda^*) : p_{ij}(F; \mathbf{G}) \geq -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma S4.3 gives $e_c(F; \mathbf{G}) \leq 16\tau$; the excess for Λ^* is the same sum with the zeroed slots dropped, so $e_c(F; \Lambda^*) \leq 16\tau < 4m^*$, and Lemma S4.2 makes F obtuse for Λ^*. □

The operators from F_0 to the members of $\mathcal{C}(\Lambda^*)$ are integer matrices independent of the metric, so one enumeration serves every band up to the one idealised at.

Lemma S4.6 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member of the closure for Z' is a member of the closure for Z. *Proof.* The members for Z' are reached from F_0 by words of flips, each at a slot that is zero at that point of the word (Lemma A3 of the main text); a flip at a slot zero for both lattices acts on both conorm vectors as the same transposition of two slots, so after each flip the superbase is obtuse for both and the next flip is again at a slot zero for both. □

Proposition S4.7 (reindexing guarantee in the reference frame). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference's reduced frame, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \leq \eta < \nu_{\min}$ with $\nu_{\min}$ the smallest eigenvalue of $\mathbf{G}_R$, let F_N be its reduced superbase with basis vectors of summed squared length V, and let $\tau \geq \sqrt{2}\,\eta\, V/(\nu_{\min} - \eta)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, with its true integer coefficients in the reference frame, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* With U the coordinates of F_N's basis vectors in the reference frame and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$,

$\sum |u_i|^2 \le V/(\nu_{\min} - \eta)$; in F_N's own frame the error has Frobenius norm at most $\sum |u_i|^2\,\eta$, and every conorm shift is at most $\sqrt{2}$ times that norm. The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem S4.5 applies. □ For fifteen references near zero conorms of five zero patterns, each with twelve new cells carrying a random metric error and a random reindexing, the true operator was among the candidates in 180 of 180 cases at the band given by the coverage bound below; the guarantee of Proposition S4.7 is conservative, a median 13η against a band near 3η that was enough.

From a metric error to a band. A metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by $-u_i^{\mathsf{T}}\Delta\mathbf{G}\,u_j$, at most $c_{ij}\eta$ with $c_{ij} = \|\operatorname{sym}(u_i u_j^{\mathsf{T}})\|_F$; every member survives every error of norm η exactly when $\tau \ge \tau_{\mathrm{cov}} = \max(c_{ij}\eta - p_{ij})$, read off the stored closure. The largest constant on zero conorms is $\sqrt{2}$ with one zero, $\sqrt{5}$ with two, and $2\sqrt{3}$ in six of the sixteen labelled cuboid frames, so $\tau = 3\eta$ is not a worst-case guarantee in every frame. These frame-dependent constants are what the strain norm removes.

S5. Further tables

Table S5. *The seven sample cells of BLDConograph (Oishi-Tomiyasu, 2024): the program's Bravais type and its minimal distance to that type (Å^2), and the same divided by the mean squared edge (%); the Bravais types of the sort spectrum in increasing strain distance (%), aP omitted, up to and including the program's type; and the next type with its distance. For all seven cells r_1 reaches the cap, so "> 25" is a lower bound on the distance as well as the outcome of the minimum over the frames of $\mathcal{U}$. A band of* 7.4% *selects the program's type in all seven; the largest ratio between consecutive distances follows it in six, and for the alumina cell lies between mC and hR.*

sample	type	Å^2	%	types up to the program's (strain %)	next
HERMES_Sr327_250K	tI	0.57	1.24	mC 0.31, oI 0.74, oF 0.83, tI 0.84	mP > 25
1339prbasrcuo_bs1	tP	0.056	0.10	mP 0.11, mC 0.14, oS 0.17, oP 0.18, tP 0.22	oF > 25
sample3	cI	2.07	0.78	mC 0.23, hR 0.39, oF 0.39, oI 0.49, tI 0.49, cI 0.58	cF > 25
CeO2_matsushita	cF	0.084	0.61	mC 0.09, oI 0.17, oF 0.17, tI 0.18, hR 0.26, cF 0.30	cI > 25
M402-pf-119673	hP	20.5	5.27	mC 0.99, mP 2.24, oS 2.37, hP 2.94	oI > 25
Acrinol-B-D8CuKa1	mC	0.72	0.36	mC 0.31	oF 20.6
Alumina676	hR	1.93	8.10	mC 0.43, hR 3.58	oI 14.9

Table S6. *The widened closure of the measured worked cell. Each row is a member, named by a word in the flips of Table 1 of the main text from the reduced cell e (the empty word); the six numbers are its conorms under the measured metric, in Å^2; "depth" is the strain at which it enters, in %. At a band of* 7.4% *all sixteen are members.*

word	*da*	*db*	*dc*	*ab*	*ac*	*bc*	depth
e	49.5	1588.9	2444.7	4.2	847.5	4.5	0
bc	45.0	8.7	2449.3	1593.4	852.0	−4.5	0.28
ab	851.7	1593.1	2440.6	−4.2	53.7	8.7	0.49
bc.db	2458.0	−8.7	53.7	1602.1	843.3	4.2	0.62
ab.bc	842.9	4.5	2449.3	1601.8	62.5	−8.7	0.78
ab.bc.db	2453.8	−4.5	847.5	1606.3	57.9	−4.2	0.78
da	−49.5	2494.3	1638.4	53.7	897.0	−45.0	3.76
ab.bc.ac	1664.2	−57.9	2511.7	905.4	−62.5	53.7	4.56
da.ab	950.7	2548.0	1584.7	−53.7	4.2	8.7	5.06
da.ab.ac	−49.5	2543.8	1588.9	954.9	−4.2	12.9	5.08
ab.ac	49.5	1539.3	2494.3	905.4	−53.7	62.5	5.08
da.ab.bc	942.0	−45.0	1593.4	2556.7	12.9	−8.7	5.09
bc.db.dc	45.0	2511.7	−53.7	1548.4	897.0	57.9	5.09
ab.ac.bc.ac	1610.5	959.1	2565.5	−4.2	−8.7	−53.7	5.11
da.ab.bc.ac	2569.6	−57.9	1606.3	954.9	−12.9	4.2	5.11
ab.ac.bc	−12.9	967.8	2556.7	1601.8	8.7	−62.5	5.11

Table S7. *The sort patterns of Andrews & Bernstein (2023), in the S^6 slot order bc, ac, ab, ad, bd, cd, and the same patterns in the slot order da, db, dc, ab, ac, bc of Table 6 of the main text. The representative used here is their pattern read in the other order, with the sign of the Selling parameters reversed; no reflection is applied. Which zeros a pattern has also fixes the kind of face: for the three-zero sorts C5, T5 and O5 the zeros are ab, ac and bc, a triangle on a, b, c, and the same lattices also have reduced cells whose zeros form a path.*

sort	S^6 order	this paper	sort	S^6 order	this paper
C1	rrrrrr	rrrrrr	O2	rs0srt	srt0sr
C3	rr0rr0	rr00rr	O3	rs0rs0	rs00sr
C5	000rrr	rrr000	O4	00rsst	sstr00
H4	00rrrs	rrsr00	O5	000rst	rst000
T1	rrsrrs	rrssrr	M1A	rrsttu	ttusrr
T2	rr0rrs	rrs0rr	M1B	rstrsu	rsutsr
T5	000rrs	rrs000	M2A	rs0stu	stu0sr
R1	rrrsss	sssrrr	M2B	rs0rst	rst0sr
R3	rr0sr0	sr00rr	M3	rs0ts0	ts00sr
O1A	rrsrrt	rrtsrr	M4	00rstu	stur00
O1B	rstrst	rsttsr	A1	rstuvw	uvwtsr
			A2	rs0tuv	tuv0sr
			A3	rs0tu0	tu00sr

Table S8. *Pseudo-symmetric PDB entries (Le Page lattice above the reported type at 3°) by the Delone sort of the Le Page lattice: number of entries, and the 10th/50th/90th percentiles of the largest obliquity δ (degrees) and of the largest and smallest price (% strain) among the pseudo-symmetry operators of each entry. Sorts with no entries are omitted. The transitions from the reported type are in Table 11 of the main text. From* `results_pdb_lepage.txt`.

sort	type	entries	δ (°)	largest price (%)	smallest price (%)
C1	cI	593	0.14/1.64/2.73	0.69/8.10/13.58	0.00/0.00/8.96
C3	cF	562	0.19/1.72/2.76	0.95/8.51/13.61	0.00/5.34/13.04
C5	cP	1 379	0.23/1.66/2.72	1.15/8.23/13.44	0.00/3.68/12.15
H4	hP	2 008	0.08/1.40/2.70	0.40/6.91/13.32	0.11/4.03/12.70
T1	tI	1 376	0.06/1.32/2.67	0.28/6.60/13.33	0.17/5.55/12.60
T2	tI	330	0.05/1.48/2.69	0.24/7.30/13.43	0.11/4.98/12.44
T5	tP	14 817	0.22/1.47/2.69	1.07/7.27/13.43	0.70/6.56/12.98
R1	hR	96	0.04/1.55/2.76	0.22/7.64/13.78	0.13/2.98/13.13
R3	hR	393	0.07/1.82/2.81	0.37/9.09/13.86	0.29/7.64/13.72
O1A	oF	1 126	0.23/1.51/2.70	1.12/7.45/13.33	1.02/7.05/13.17
O1B	oI	331	0.10/1.58/2.78	0.49/7.80/13.73	0.36/7.63/13.73
O2	oI	1 078	0.14/1.40/2.73	0.70/6.92/13.50	0.48/6.33/13.36
O4	oS	11 679	0.14/1.49/2.74	0.67/7.36/13.56	0.62/7.15/13.51
O5	oP	4 760	0.03/1.08/2.59	0.17/5.33/12.79	0.15/5.13/12.69
M1A	mC	867	0.72/2.09/2.83	3.53/10.33/13.98	3.53/10.33/13.98
M1B	mC	398	0.12/1.80/2.72	0.57/8.91/13.41	0.57/8.91/13.41
M2A	mC	854	0.13/1.75/2.77	0.66/8.64/13.66	0.66/8.64/13.66
M2B	mC	2	0.61/0.95/1.29	3.02/4.70/6.38	3.02/4.70/6.38
M3	mC	1	2.39/2.39/2.39	11.80/11.80/11.80	11.80/11.80/11.80
M4	mP	582	0.10/1.40/2.65	0.47/6.89/13.07	0.47/6.89/13.07

Over the sorts with more than two entries, the median largest price lies between 5.3% (O5, oP) and 10.3% (M1A); the median largest obliquity between 1.1° and 2.1°.

Within a sort the median largest price of the operators of order 3, 4 or 6 is within half a percentage point of that of the twofolds (`results_pdb_lepage.txt`).

S6. Computational checks

Each entry of Table S9 names the test files that reproduce the numbers of a section, table or figure of the main text. A mark *znn* is `manuscript/appendix_z/test_z`*nn*`_*.py` in the `agentsg` repository, lib is `agentsg/tests/test_strain_closure.py`, and bench names a script in `manuscript/benchmark`. The checks were run with Python 3.13.16, NumPy 2.5.3 and SciPy 1.18.1, and the comparisons with cctbx, SELLA (LatticeRepLib) and the NCDist/CS6Dist libraries; the checks z24, z25 and z42 are skipped when their external data are absent, and none of the numbers of the paper rests on them; the PDB scripts need the Zenodo snapshot (doi:10.5281/zenodo.22986222) in `data/`.

Table S9. *The checks behind each part of the main text.*

main text	checks
§2.1 faces; §2.2 Table 1	z52, z56 (the 38 faces); z46 (flip matrices)
§3.1 counts of superbases and objects	z54, z56
§3.2, Table 3, Fig. 1	z46 (the ring, the action of $2/m$, the word *ab.bc.db*)
§3.3 $\mathcal{U}$	z52 (6096, inversion, window, nine kinds), z56 (every window member in $\mathcal{U}$; about 25 min)
§4.1, Table 4	z60 (twofold price from obliquity, the 92° cell)
§4.2–4.3	z59 (balls, layers of 127 and 2755, filtered layer), z60 (Lemma A8); z44, z45 (difference-wall members, walks)
§4.4–4.5	lib (reindexing, scale bound, volume check, classes), z60 (Proposition A10)
§4.7 windows	z52, z56 ($[-2, 2]$ labelling), z58, z59 (brute force)
§5.1, Table 6	z53, z54, z46, z49 (Table S1), z50 (Table S2)
§5.2, Table 7	z56 (entries by zero pattern), z60 (the Niggli type decides zeros on d near oP); bench: `niggli_types.py`, `window_failures.py`
§5.3, Fig. 2, Table 8	z60 (depths, growth of the set), z57, z58, z59
§5.4, Tables 9–10, Fig. 3	lib, z60 (classes, prices, centred references); bench: `evaluate_strain.py`, `compare_methods.py`, `window_failures.py`, `coset_rank.py`
§5.5, Fig. 4, Table 11, Table S8	bench: `pdb_nonisomorphism.py`, `pdb_lepage.py`, `pdb_pricing_summary.py` (need the Zenodo snapshot)
Appendix A	z56 (A1), z44 (A2), z53 (A5), z59 (A6, A7), z60 (A8, A9, A10)
Appendix B, Fig. B1, Table B1, Fig. S1, Table S5	z58, z60, z51; bench: `lepage_calibration.py`, `compare_sella.py`
§S4	z36, z44 (Lemma S4.4), z45, z46, z47, z55, z57

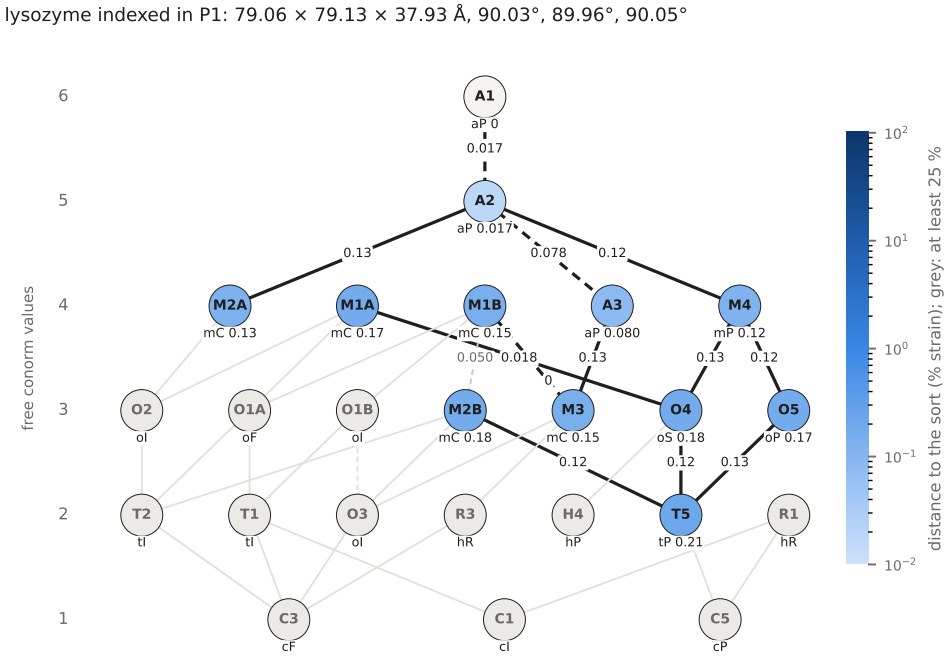


Fig. S1. The 24 Delone sorts as a graph for a tetragonal lysozyme cell (79.1 × 79.1 × 37.9 Å) indexed in $P1$ with an added error, drawn as Fig. B1 of the main text. The twelve sorts from aP to tP lie within 0.22%; every other sort is at least 25% away (grey).

References

Andrews, L. C. & Bernstein, H. J. (2023). *Acta Cryst.* A**79**, 499–503.

Andrews, L. C., Bernstein, H. J. & Pelletier, G. A. (1980). *Acta Cryst.* A**36**, 248–252.

Conway, J. H. & Sloane, N. J. A. (1992). *Proc. R. Soc. London Ser. A*, **436**, 55–68.

Delaunay, B. (1933). *Z. Kristallogr.* **84**, 109–149.

Hahn, Th. (2006). Editor. *International Tables for Crystallography*, Vol. A, 5th ed. Dordrecht: Springer.

Oishi-Tomiyasu, R. (2024). *Acta Cryst.* A**80**, 339–350.

Selling, E. (1874). *J. Reine Angew. Math.* **77**, 143–229.